Akoizo Original Practice - Form 5

Division C | Electric Vehicle | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which pusher feature is NOT allowed?

Unit: The bottle pusher: catching, pushing, and the narrow-opening bonus.
Worked-case data: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).
Separate investigation observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

A. Hinged arms that close around the bottle

B. A rigid V-shaped funnel

C. Rounded front edges

D. A curved pocket

Answer: _____

2. (2 points) Why are quadrature encoders useful in this event?

Unit: Drivetrain and distance sensing: motors, gears, and encoders.
Worked-case data: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?
Separate investigation observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

A. They detect direction, which matters when the vehicle reverses

B. They measure battery voltage

C. They replace the pencil trigger

D. They measure the bottle distance

Answer: _____

3. (2 points) The bottle never gets past the Target Point. What Bottle Distance is used?

A. 400.0 cm

B. 0 cm

C. The measured distance

D. 2500 cm

Answer: _____

4. (2 points) A team uses a kit with only one documented modification. What happens?

A. No penalty

B. A 300-point construction violation

C. A 50-point non-modification penalty on its runs

D. Participation points only

Answer: _____

5. (4 points) Why is closed-loop speed control better than fixed power for timing?

Answer: _____
_____
_____

6. (4 points) Why should the program take the posted values as inputs rather than hard-code them?

Answer: _____
_____
_____

7. (4 points) Why does the supervisor pause event time before each run?

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

Answer: _____
_____
_____

8. (4 points) Why push low on the bottle?

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Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
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Answer: _____
_____
_____

9. (4 points) Why choose a gear ratio with speed margin?

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Separate investigation observations:
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Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: Target Time 14 s. Pusher opening 16.0 cm. The vehicle stops 22.5 cm from the Target Point in 16.37 s; the bottle ends 12.1 cm beyond the Bottle Line. Find the Run Score.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A team’s vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?

Answer: _____
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_____

12. (2 points) What does the comparison show?

Supplied synthetic observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

A. Feedback control makes time and stopping independent of battery state

B. Fresh batteries are always slower

C. Fixed power is more accurate

Answer: _____

13. (2 points) Define “Target Distance” in the context of Electric Vehicle.

Answer: _____
_____
_____

14. (2 points) Define “Impound” in the context of Electric Vehicle.

Answer: _____
_____
_____

15. (2 points) Define “Widest opening” in the context of Electric Vehicle.

Answer: _____
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16. (2 points) Define “Quadrature” in the context of Electric Vehicle.

Answer: _____
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17. (2 points) Define “Bottle Bonus” in the context of Electric Vehicle.

Answer: _____
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18. (2 points) Define “Single-source program” in the context of Electric Vehicle.

Answer: _____
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